

Cross-border platforms' coupling of artificial intelligence adoption and strategic agility for upgraded ESMEs' global value chains: international marketing capability perspectives

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Abstract

Purpose – Despite a broad consensus on firm internationalization through digital platforms, it still appears to be a black box mechanism through which emerging market small and medium-sized enterprises (ESMEs) upgrade their global value chain (GVC) position. This study aims to investigate how artificial intelligence (AI) adoption and strategic agility coupling shape ESMEs' GVC positioning through the lens of international marketing capabilities (IMCs), differentiating international advantage into two dimensions: product innovation and brand upgrading. Furthermore, it investigates how the depth and breadth of platform embeddedness moderate this relationship.

Design/methodology/approach – A quantitative approach was employed to analyse data from 322 small and medium-sized enterprises in China utilizing cross-border platforms. The research model was tested using structural equation modelling, with data analysis conducted through SPSS 27, AMOS 24 and STATA to evaluate both the measurement and structural models.

Findings – The study demonstrates that the coupling of AI adoption and strategic agility positively influences IMCs, product innovation and brand upgrading, while IMCs serve as a mediating mechanism. Additionally, platform embedding characteristics exhibit a nonlinear, inverted U-shaped moderating effect: a medium level of embedding depth maximizes the coupling's positive impact on product innovation, whereas a moderate level of embedding breadth optimizes its positive effect on brand upgrading.

Originality/value – This study proposes a holistic framework that explores how platform affordances facilitate the identification and forecasting of international market information and resource redeployment within platform ecosystems, driving product and brand value creation. It advances the dynamic capabilities hierarchy in international business research by integrating digital platforms and the ESMEs perspective, illustrating how higher-order strategic agility shapes embedded ESMEs' lower-order IMCs, enhancing international competitive advantage. Additionally, it examines AI-enabled marketing capabilities through a broader lens of international advantage.

Keywords Artificial intelligence adoption, Strategic agility, International marketing capabilities, Global value chains, Cross-border platforms, ESMEs

Paper type Research article

1. Introduction

International business research is currently propelled by the phenomenon of “platformization”, which represents an open interface between different users that facilitates value-creating



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exchanges through the use of digital technologies (Cennamo and Santalo, 2013; Nambisan *et al.*, 2019; Kim and Cavusgil, 2020). Typically, cross-border platforms act as key intermediaries, reducing distance barriers and expanding access to firms that interact with customers, suppliers and stakeholders (Li *et al.*, 2019; Gong *et al.*, 2024; Teece, 2025). The rise of digital technologies is reshaping global business operations derived from intangible assets and data, alongside instant access to global knowledge. Cross-border platforms' AI adoption – incorporation of the programs, algorithms, systems and machines that emulate human cognitive functions to identity and core processes – fundamentally impacts the international knowledge generation, analysis and decision-making process and act upon real-time (Huang and Rust, 2021; Liu *et al.*, 2024). For emerging economies small and medium-sized enterprises (ESMEs) with limited resources, brand recognition or technological access rely on platform-based AI adoption to acquire strategic assets, restructure international marketing, as well as enhance core functions such as R&D and brand management (Nambisan *et al.*, 2019; Xu *et al.*, 2023). Consequently, cross-border platforms function not only as multisided marketplaces but also as strategic venues for ESMEs. Existing research has examined the export performance, internationalization speed and scope outcomes of firm internationalization using digital platforms (Luo and Bu, 2018; Kim and Cavusgil, 2020; Liu *et al.*, 2024). A gap persists in understanding their role in post-entry global value chain (GVC) position upgrading for ESMEs.

Upgrading GVCs to achieve sustainable development increasingly relies on advanced infrastructure adoption, such as AI (Hossain *et al.*, 2022; Da Rocha *et al.*, 2024). In line with this, the UN's Sustainable Development Goals (SDGs) emphasize building resilient technological infrastructure to promote global sustainability (UN, 2015). Such as buyer–seller interaction toolkits, sponsored search listings and personalized recommendation algorithms, supporting ESMEs' sustainability goals without incurring development costs. These AI-enabled tools enhance global brand and product positioning and generate market insights that help ESMEs bridge structural gaps across borders and deliver solutions in marketing, innovation and online word-of-mouth (Raisch and Krakowski, 2021; Manis and Madhavaram, 2023; Ameen *et al.*, 2024). Despite these advancements, scholarly understanding of how AI adoption impacts internationalization advantage has not kept pace. Recently, the internationalization literature has begun to explore the role of AI strategy and capability in facilitating the internationalization speed (e.g. Liu *et al.*, 2024). However, existing research has yet to explore how platforms integrate AI technologies to facilitate ESMEs' GVC upgrading. This study aims to offer a pragmatic approach by distinguishing international advantages into two dimensions: product innovation and brand upgrading.

While AI adoption increases the availability of international market information, its value lies in the strategic agility to continuously adapt AI-enabled insight within platform-based ecosystems. This coupling enables ESMEs to maintain a dynamic balance between innovative products and brands and respond to AI-driven opportunities (Demir *et al.*, 2021; Ahammad *et al.*, 2021). Considering Teece's elaboration on strategic agility – as the capacity to redeploy its resources to value-creating and capturing higher-level activities as environments warrant (Teece *et al.*, 2016), this paper clarifies strategic agility as a platform's higher-order dynamic capability to identify, assemble and orchestrate profitable configurations of competencies and assets, thereby maximizing AI's potential (Schoemaker *et al.*, 2018; Shams *et al.*, 2021). While prior research has often focused on how AI adoption advances firm innovation and competitiveness (e.g. Davenport *et al.*, 2020; Garbuio and Lin, 2021; Hossain *et al.*, 2022) or how strategic agility enhances internationalization speed or success separately (e.g. Demir *et al.*, 2021; Ahammad *et al.*, 2021; Ferraris *et al.*, 2022), it neglects how their coupling shapes the heterogeneity of ESMEs' international advantage. Thus, more in-depth research is needed on how the coupling of AI adoption and strategic agility empowers ESMEs to address near-future international market opportunities and develop viable product and brand activities to overcome “low-end lock-in” constraints.

ESMEs typically invest heavily in product and brand development, yet the absence of effective marketing capabilities can trap them in an ineffective cycle (Chatterjee *et al.*, 2022; Mostafiz *et al.*, 2024). As Zahoor and Lew (2023) argue, international marketing capabilities (IMCs) can facilitate price adjustment, product development and deploy tailored marketing strategies for foreign markets. According to Schoemaker *et al.* (2018), lower-level operational capabilities underpin basic functional activities of products and brands, while dynamic capabilities help identify new products and brand positioning, potentially opening new markets where rivals have not yet appeared. Thus, we aim to draw on the hierarchy view of dynamic capability (Zollo and Winter, 2002; Winter, 2003), investigating how cross-border platforms' AI adoption, coupled with higher-order dynamic capabilities, systematically alter ESMEs' lower-level IMCs, ultimately upgrading their GVC positions.

Recent studies show that AI-enabled marketing capabilities enhance various functions, such as analysing competing products, designing based on user feedback, improving market sensing and optimizing distribution channels. Existing studies primarily examine IMCs in export marketing performance or early-stage internationalization (Wang, 2020; Manis and Madhavaram, 2023; Pfajfar *et al.*, 2024a; Mostafiz *et al.*, 2024). Despite growing interest in AI-enabled marketing capabilities, for example, Hossain *et al.* (2022) revealed that firms' marketing analytics capability, when enhanced by AI, significantly improves their ability to seize market opportunities. However, the conceptual and empirical approaches to AI-enabled IMCs' role in post-entry international advantage remain underexplored. Hence, examining IMCs as a mechanism can deepen our understanding of how AI adoption, coupled with strategic agility, drives ESMEs' product innovation and brand upgrading.

Despite the growing recognition of platforms' importance in firms' internationalization, research has primarily focused on factors driving firms' internationalization via digital platforms, emphasizing firm-specific resources, capabilities, platform types, industry conditions and country-specific contexts (Da Rocha *et al.*, 2024; Gong *et al.*, 2024). Additionally, studies have mainly examined platform participation in outcomes such as foreign market entry, overseas sales, international scope, intensity and financial performance (e.g. Demir *et al.*, 2021; Zahoor and Lew, 2023). However, while many ESMEs are advancing internationally without achieving global status, significant gaps remain in understanding how they engage with cross-border platforms, specifically in leveraging AI and strategic agility to drive product innovation and brand upgrading, despite the recognized importance of GVC upgrading. Moreover, we suggest that the magnitude of this change may hinge on ESMEs' embedding characteristics. The emergence of ESMEs' internationalization through cross-border platforms introduces two characteristics: embedding breadth and depth. Breadth refers to the variety of platform types engaged (e.g. multi-homing), whereas depth captures the intensity of interactions with platforms (Li *et al.*, 2019; Kobarg *et al.*, 2019). As a result, these embedding differences may lead to differential quality and quantity of information and resources accessible to ESMEs through digital platforms. Accordingly, this study seeks to answer the following research questions to illuminate the dynamic interplay between ESMEs and cross-border platforms and to maximize competitive gains:

- (1) How does the coupling of AI adoption and strategic agility within a cross-border platform enable GVC upgrading for ESMEs?
- (2) What role do the IMCs of ESMEs play in this process?
- (3) How do platform embedding breadth and depth separately moderate the effect of coupling AI adoption with strategic agility on product innovation and brand upgrading?

To investigate the above research question, we performed a large-scale survey of Chinese small and medium-sized enterprises. China's cross-border e-commerce unleashes its potential in "selling globally", reaching 2.63 trillion RMB in 2024, a 10.8% year-on-year increase.

Cross-border platforms such as AliExpress, TikTok and Shein have adopted AI technologies to help Chinese SMEs mitigate foreign market entry risks, increase global clients, reduce transaction costs and bolster brand visibility (Li *et al.*, 2022; Da Rocha *et al.*, 2024). As a result, Chinese SMEs are evolving from low-end assemblers to high-value contributors in the GVC, such as R&D and brand building. Hence, the shifting positions of ESMEs in GVCs through cross-border platforms serve as an ideal context for our purpose. To capture the causal effects of the coupling of AI adoption and strategic agility on ESMEs' international competitive advantage, we conducted a questionnaire-based survey of Chinese SMEs embedded in cross-border platforms such as Alibaba International Station, Global Express, Amazon, TikTok, Temu and SHEIN to drive product innovation and brand upgrading.

This study makes four key contributions. *First*, our research offers novel insights into the role of digital platforms in international business by proposing a novel coupling framework linking AI adoption and strategic agility (e.g. Jean *et al.*, 2020; Da Rocha *et al.*, 2024). This coupling framework extends prior work that has typically examined AI and strategic agility separately. In doing so, unlike extant research, which explained that the use of digital platforms can affect internationalization speed, scope and performance (Coviello *et al.*, 2017; Li *et al.*, 2019; Gong *et al.*, 2024), our research reveals how ESMEs post-entry internationalization upgrade their GVC positions using digital platforms. *Second*, it contributes to the literature on strategic agility by combining this agility with AI adoption can lead to improved product innovation and brand upgrading (Ahmad *et al.*, 2021). This study conceptualizes strategic agility not only as a high-level dynamic capability but also as a response to AI-driven predictive insights, enhancing international market strategic foresight, reconfiguring resources and driving value creation in volatile global markets. *Third*, this study extends the dynamic capabilities hierarchy in international business (Winter, 2003; Teece *et al.*, 2016) by adopting an interactive digital platform–firm perspective. By providing empirical insight into higher-order dynamic capabilities that drive “regular” operational capabilities for GVC upgrading (Eng and Spickett-Jones, 2009), this study extends AI-driven marketing capabilities to international business research, particularly for ESMEs in advanced internationalization that have not yet attained global corporate status. *Finally*, this study further shows that the coupling of AI adoption and strategy affects the international advantage of products and brands, depending on ESMEs' platform embedding characteristics, highlighting critical yet underexplored moderating factors beyond institutional voids, market uncertainty and geographical distance (Ko *et al.*, 2021).

This paper proceeds as follows: Related work and theoretical background are presented in Section 2. Section 3 elucidated the hypotheses and conceptual model development. Section 4 describes the data collected and the empirical strategy. Section 5 presents the study's empirical analysis and results. Section 6 discusses our findings and provides theoretical and practical implications for researchers, policymakers and practitioners. Section 7 concludes the paper and acknowledges some of the limitations and prospects of our study.

2. Theoretical background

2.1 Coupling AI adoption and strategic agility

AI adoption is defined as the incorporation of technology into an organization's identity, integrating machines' ability to identify, interpret, infer and learn from data to create flexible and accurate decisions (Mikalef and Gupta, 2021; Dahlke *et al.*, 2024). In the context of ESMEs' internationalization via digital platforms, cross-border platforms' investment in AI infrastructure enables ESMEs to overcome traditional infrastructure constraints (Luo and Bu, 2018; Wei and Pardo, 2022). For example, Ahi *et al.* (2022) highlight that advanced technologies have important implications for firms that govern the structure of their value chains through digital platform networks. By adopting AI, platforms can generate, analyse and act upon real-time data at unprecedented speed and scale (Massa *et al.*, 2023; Liu *et al.*, 2025). Thus, the cross-border platform not only provides ESMEs with access to foreign networks for

global expansion but also offers a range of advantages through AI adoption, supporting their international competitiveness (Hossain *et al.*, 2022; Manis and Madhavaram, 2023; Dahlke *et al.*, 2024). For example, platforms' AI adoption allows more value to be created by offering cognitive analyses of global consumer behaviour, mining global user profiles through online footprints and user feedback, facilitating real-time multilingual communication and streamlining logistics and payment systems to its participants (Wei and Pardo, 2022). Accordingly, ESMEs embedded in the platform constantly reshape their international competitive advantages by mitigating challenges, such as language barriers, temporal and spatial limitations in gathering international market insights, cross-border payment issues and logistical complexities.

Cross-border platforms are profoundly influenced by AI, with agile platforms better positioned to leverage these AI technologies for rapid market responsiveness and operational efficiency and develop customer-centric solutions (Kano, 2018; Ahammad *et al.*, 2021; Ahi *et al.*, 2022; Abdalla *et al.*, 2025). Grounded in the dynamic capability perspective, strategic agility is a high-order dynamic capability, which refers to the platform's ability to adapt and reconfigure its resources to respond effectively to AI-driven international market insights (Teece *et al.*, 1997; Winter, 2003; Ahammad *et al.*, 2021). AI adoption supports this strategic agility by reducing information asymmetry, enabling predictive insights and facilitating decentralized decision-making (Liu *et al.*, 2025). In particular, ESMEs in weaker institutional environments, institutional pressures and limited competitive advantages constrain agility (Luo and Bu, 2018; Abdalla *et al.*, 2025), underscoring how the platform's strategic agility mitigates inherent resource limitations (Ahammad *et al.*, 2021; Zahoor and Lew, 2023). According to a study by Shams *et al.* (2021), strategic agility is a key intangible asset in international business. Similarly, Fourné *et al.* (2014) describe it as a meta-capability that allows companies to swiftly identify local opportunities, favour global complementarities and promote sustainability. Furthermore, Abdalla *et al.* (2025) argue that by embracing agility, platforms can integrate emerging technologies. Strategic agility also supports the global brand development identity by enabling firms to respond quickly to user feedback and strengthen their reputation (Tallon and Pinsonneault, 2011; Demir *et al.*, 2021; Mancuso *et al.*, 2024). Notwithstanding its recognized importance, there remains a notable gap in understanding how ESMEs in weak institutional settings leverage cross-border platforms' strategic agility to advance in GVCs.

As global attention to environmental, social and governance standards continues to grow, underscoring the need for the ESME to explore strategies for achieving global sustainable development (Luo and Bu, 2018; UN, 2015). Given their resource limitations, exposure to international competitive pressures and institutional pressures often complicate this process. The majority of studies have focused on how digital platforms enable SMEs to acquire foreign knowledge, develop international opportunities, reduce export-related costs and lower entry barriers, thereby influencing the scope and performance of internationalization (e.g. Jean *et al.*, 2020; Da Rocha *et al.*, 2024). While existing research confirms the role of AI adoption in enhancing performance, innovation and competitive advantage (e.g. Gama and Magistretti, 2023; Raisch and Krakowski, 2021), little is known about how AI adoption on platforms evolves in international business. Furthermore, prior research emphasizes the standalone effects of AI adoption or strategic agility and few studies examine how these constructs are coupled within a unified framework. This theoretical gap is particularly salient given the complex nature of ESMEs navigating complex GVC upgrading, where agility must be complemented by robust informational infrastructures.

2.2 International marketing capabilities (IMCs)

IMCs are defined as a firm's ability to deliver products that meet the needs of targeted market segments and to deploy marketing resources effectively in turbulent foreign markets (Eng and Spickett-Jones, 2009; Fang and Zou, 2009; Zahoor and Lew, 2023). IMCs are conceptually

linked to shifts in international marketing strategies (Fang and Zou, 2009). Accordingly, for resource-limited ESMEs, cultivating robust IMCs is critical for securing competitive positions in GVCs. Since AI analytics provide data-driven knowledge, they offer accurate and reliable market information, enabling firms to flexibly respond to international market opportunities that might be overlooked by humans. AI-enabled cross-border platforms thus enhance the ESME's ability to marketing research, strategy and action stages – key tenets of the IMCs framework (Huang and Rust, 2021). For example, at the marketing research stage, AI can be effectively employed for data collection, in-depth market analysis and customer understanding. Progressing to the marketing strategy phase, AI facilitates precise segmentation recognition, targeting and positioning. At the marketing action stage, AI contributes to standardization, personalization and relationalization. This structured application of AI across different marketing stages underscores its pivotal role in optimizing marketing efficacy and strategic alignment. Despite emerging research on AI-enabled marketing capabilities (Huang and Rust, 2021; Manis and Madhavaram, 2023; Zahoor and Lew, 2023), international marketing scholarship has inadequately examined their role in global international contexts.

From a hierarchical view of dynamic capability, successful international development necessitates higher-order dynamic capabilities for adapting to the global market environment and adjusting resource bases. AI operations, in practice, function synergistically with other resources to enhance ESMEs' capabilities, including diversified marketing tools that optimize market capabilities (Dahlke *et al.*, 2024). IMCs serve as vital lower-order operational capabilities that support various aspects of core marketing activities (Winter, 2003). In this setting, ordinary capabilities are the ESME's international marketing routines (operationalized in product or brand), and we study the processes through which such routines change, affected by platforms' coupling technical infrastructure and higher-order dynamic capabilities. In this regard, ESMEs need to utilize platforms' higher-order dynamic capabilities for developing lower-order IMCs that address market-specific demands, such as market conditions, consumer preferences, cultural nuances, regulatory environments and distribution channels embedded in market practices, thereby effectively delivering value in GVCs (Teece *et al.*, 2016; Ahammad *et al.*, 2021; Schulze and Brusoni, 2022). Platforms exhibiting superior strategic agility can co-create value with participants, highlighting the importance of AI adoption for ESMEs to identify market demands and emerging opportunities (Pfajfar *et al.*, 2024b). Thereby, the coupling of AI adoption and strategic agility allows ESMEs to extend, modify or create ordinary capabilities, allowing them to respond faster to shifting market demands and capitalize on emerging opportunities (Winter, 2003). Accordingly, as lower-order operational capabilities, IMCs translate customer knowledge into tactical marketing applications, enabling product innovation and brand development.

Empirical studies have consistently revealed a well-supported relationship between IMCs and firms' international performance outcomes, such as financial metrics, export results and innovation performance (e.g. Eng and Spickett-Jones, 2009; Chatterjee *et al.*, 2022). However, the results can vary across different contexts and samples. Nonetheless, the international marketing literature still lacks more granular advantages research, including new product innovation and global brand upgrading (Pfajfar *et al.*, 2024b). Furthermore, research has typically examined IMCs in isolation without adequately accounting for their mediating roles. This lack of integration into the interaction perspective between digital platforms and ESMEs limits the explanatory power of existing theories in explaining international competitive advantage heterogeneity among ESMEs via cross-border platforms. Addressing these questions could help build a richer theoretical understanding of how coupling AI adoption and strategic agility enables ESMEs to upgrade their GVC position.

2.3 Global value chains upgrading of ESMEs as a research context

Rapid shifts in technological innovation, market upgrading, GVC restructuring, and geopolitical tensions are causing traditional emerging economies small and medium-sized

enterprises (ESMEs) to lose their competitive edge in low-cost, large-scale production (Adomako *et al.*, 2019; Gölgeci *et al.*, 2023). ESMEs now face both opportunities and challenges in upgrading their GVC to reshape their international advantages. GVCs consist of interconnected processes in the production, distribution and consumption of goods and services worldwide; thus, product innovation and brand upgrading are critical for enhancing value chain positioning (Caputo *et al.*, 2022; Gölgeci *et al.*, 2023). Unlike SMEs in developed economies, ESMEs often suffer from institutional voids and limited resources (Ko *et al.*, 2021; Singh *et al.*, 2022), which can trap them in a “low-end lock-in”, resulting in low-profit margins within GVCs (Adomako *et al.*, 2019). The development of cross-border platforms should not only foster growth points and new momentum for foreign trade but also foster independent brands and innovative development momentum for enterprises. Consequently, ESMEs compete not based emphatically on the institutional and policy environment and resources shortage, etc. in which they may be disadvantageous, but on the creative utilization and combination of internal and external resources of the cross-border platforms in which they are embedded to enhance their international value-added activities, such as producing high-quality, branded products.

As one of the largest cross-border e-commerce markets, China’s cross-border e-commerce exports exceeded 1.9 trillion yuan in 2024, marking a 15.2% increase and underscoring the sector’s pivotal role in emerging economies and global trade growth. The dynamic nature of the Chinese market presents a valuable context for investigating how cross-border platforms contribute to supporting the potential of product innovation and brand upgrades for ESMEs. However, current research predominantly focuses on the impact of digital platforms on international performance and scope, with the literature largely centred on firms from developed countries. This narrow focus offers limited insights into how ESMEs can leverage their limited resources and weak formal institutions to strengthen their international standing (Da Rocha *et al.*, 2024). We believe that AI adoption and strategic agility coupling models have great potential to explain why and how some ESMEs without possessing traditionally defined monopolistic advantages such as global market power, brand images, organizational reputation, customer loyalty and proprietary technologies can upgrade their position in the GVC. Addressing this gap holds both theoretical and practical significance in understanding how cross-border platforms can reposition ESMEs from low-end participants to key innovators and brand leaders (Gölgeci *et al.*, 2023).

3. Hypothesis and conceptual model development

3.1 *Coupling of AI adoption and strategic agility for international competitive advantage*

We propose that coupling AI adoption with strategic agility creates significant opportunities for ESMEs to achieve international competitive advantage. AI adoption includes routines for data collection and analysis, supporting data-driven decision-making and crafting innovative scenarios (Verganti *et al.*, 2020; Garbuio and Lin, 2021). Specifically, AI’s capabilities for sensing, interpreting and evaluating market data can act as powerful tools for product creativity and brand enhancement through automation and augmentation (Raisch and Krakowski, 2021; Ameen *et al.*, 2024). For example, generative AI technologies like Open AI’s system facilitate the expansion of both problem and solution spaces, driving product and brand advantages in alignment with design thinking principles.

Product innovation has long been a primary avenue for establishing international competitive advantage, particularly by developing or enhancing products that offer unique benefits and superior quality compared to those of competitors (Cui and Wu, 2017; Hurtado-Torres *et al.*, 2018). Cross-border digital platforms deploy AI to provide real-time data across the value chains; this facilitates collaboration among participants, shortens experimentation cycles and speeds up international market entry (Mariani and Nambisan, 2021; Gama and Magistretti, 2023; Del Giudice *et al.*, 2023). AI on these platforms also helps aggregate and identify diverse consumer demands, enhancing the accuracy of market predictions and

empowering ESMEs to bridge cultural and geographical divides, thereby fueling product innovation (Garbuio and Lin, 2021; Verganti *et al.*, 2020). Meanwhile, brand upgrading refers to the ability of export firms to achieve higher customer cognition, satisfaction and loyalty toward their products compared to foreign competitors (Lang *et al.*, 2023; Zahoor *et al.*, 2024). When brands operate across different cultures, managing value perceptions becomes more complex (Spyropoulou *et al.*, 2011). Here, AI assists ESMEs by supporting targeted advertising and automatically personalised content that resonates with specific cultural and institutional contexts. This approach facilitates the precise communication of brand image, improves the international exposure and dynamic stakeholders' interactions, reduces the difficulty of cross-cultural content adaptation and ultimately increases international brand recognition and customer loyalty (Spyropoulou *et al.*, 2011; Mandler *et al.*, 2021). For example, AI-driven chatbots offer real-time engagement across time zones, which Mandler *et al.* (2021) argue builds trust and credibility, strengthening brands' market positions abroad.

Strategic agility is particularly crucial for ESMEs' internationalization efforts due to the variable institutional support and rapid, unpredictable changes in global markets (Adomako *et al.*, 2019; Ahammad *et al.*, 2021). It entails the timely sensing of opportunities, rapid decision-making and adaptive strategic direction adjustments, which are essential for managing disruptive innovations and navigating unpredictable market conditions (Fourné *et al.*, 2014; Zahoor *et al.*, 2024). Given that ESMEs often lack strong brand recognition or cutting-edge technology, we argue that platforms' strategic agility allows these ESMEs to leverage platform resources to reconfigure their existing resource base, enabling swift responses to evolving market trends (Ahammad *et al.*, 2021). This flexibility empowers ESMEs to bridge competitive gaps, particularly by swiftly aligning resources with international market demands through digital platforms to enable new product development and enhance brand recognition (Ameen *et al.*, 2024). The literature has addressed the role of strategic agility in international business or international performance (e.g. Fourné *et al.*, 2014; Demir *et al.*, 2021), though mainly focusing on its isolated effects rather than examining its potential synergy with AI. AI supports this agility by enabling automated and data-driven decision-making processes, which are particularly valuable for product innovation and brand positioning (Raisch and Krakowski, 2021; Gama and Magistretti, 2023). Recent studies comparing optimal profits with and without AI have found that manufacturers' optimal profits in both modes consistently increase with platform power when AI is integrated (Xu *et al.*, 2023). Despite these advancements, limited research has explored the combined impact of AI adoption and strategic agility on driving product innovation and brand advantage. Thus, we formulate the following hypothesis:

- H1. The coupling of AI adoption and strategic agility has a significant positive effect on (a) product innovation advantage and (b) brand upgrading advantage.

3.2 Direct and mediating effects of IMCs

IMCs refer to the ability of ESMEs to reconfigure marketing resources, effectively identify foreign marketing needs, introduce new products to meet these needs and deliver value in foreign markets, ultimately helping ESMEs create value in the export market (Papa *et al.*, 2022; Zahoor and Lew, 2023). Innovation performance has been extensively observed by studies on a sample of SMEs operating in various foreign markets; evidently, SMEs with IMCs are becoming more innovative in international markets (Pfajfar *et al.*, 2024b). For example, early on, Eng and Spickett-Jones (2009) indicate that marketing capabilities help Chinese companies manufacture upgrades in international value chains. A similar mechanism recently revealed indicates that the focal marketing capabilities developed by born-global firms help in addressing a global market gap or being first to market with radical, patented product innovations (Zahoor and Lew, 2023; Pfajfar *et al.*, 2024b). Thus, we maintain that IMCs can engender product innovation advantages for ESMEs.

Moreover, ESMEs that possess IMCs can realise brand advantage, transitioning from original equipment manufacturing to original design manufacturing and, ultimately, to own-brand manufacturing (Eng and Spickett-Jones, 2009). In the area of international branding, IMCs encompass specific categories, such as marketing communications, brand management, market information acquisition, interpretation and dissemination, all of which are expected to influence foreign customers' mental responses towards the brand (Chatterjee *et al.*, 2022; Pfajfar *et al.*, 2024b). For example, market and process knowledge, along with export market experience, enables firms to effectively adapt their brands to local characteristics (Liu *et al.*, 2025; Zahoor *et al.*, 2024). Notably, managing customers' brand perceptions is particularly complex, requiring brands to tailor their positioning to fit local market characteristics. Spyropoulou *et al.* (2011) indicate that an exporting firm's communication capability, through its market sensing and linking customer attitudes and behaviour, significantly enhances its branding advantage. Collectively, ESMEs with superior IMCs may strengthen their brands' global positioning. On these grounds, we hypothesise:

- H2. IMCs have a significant positive effect on (a) product innovation advantage and (b) brand upgrading advantage.

Drawing further on the hierarchical view of dynamic capability (Winter, 2003), we aim to elucidate that the coupling of AI adoption and strategic agility directly influences IMCs, which, in turn, mediate the impact of this coupling on product innovation and brand upgrading advantages. According to Winter (2003), ordinary capabilities enable short-term operational success, while dynamic capabilities modify, extend or create these ordinary capabilities. Previous research argues that higher-order dynamic capabilities, like strategic agility, do not directly enhance competitive advantage but do so by shaping lower-order operational capabilities (Fourné *et al.*, 2014; Ahammad *et al.*, 2021). In this framework, strategic agility acts as a higher-order dynamic capability, coupling with AI adoption to facilitate the development of lower-order IMCs, which then strengthen ESMEs' international competitive advantage.

The coupling of AI adoption with strategic agility offers significant potential for enhancing IMCs. AI can reinforce IMCs by enabling strategic agility to combine or integrate with customers, partners, global supply chain and other resources, regardless of geographical boundaries (Hossain *et al.*, 2022; Manis and Madhavaram, 2023). AI adoption are also becoming standard for cross-border platforms, marketing functions and activities that can benefit from AI, due to its ability to recognize patterns from data (e.g. text mining, speech recognition, facial recognition) (Huang and Rust, 2021). Specifically, AI can collect and synthesise vast amounts of global user data from various international markets, while advanced cognitive computing transforms tacit knowledge into segmentation insights, enabling ESMEs to uncover latent market positioning and gain a more comprehensive understanding of market dynamics within their industry (Manis and Madhavaram, 2023). For instance, Alibaba International Station's AI-powered risk control agent can swiftly identify market risks in response to rapidly evolving global trade policies and product standards. Notably, when cross-border platforms demonstrate strategic flexibility, they can flexibly reallocate and reconfigure global resources and competencies (Zahoor *et al.*, 2024). In this regard, ESMEs enable leveraging platforms' powerful resource advantages, such as intelligent supply chains, to expand market coverage and enhance IMCs, allowing them to quickly adapt to varying market conditions across different countries and regions (Zahoor and Lew, 2023). Thus, although, being resource-constrained, ESMEs may not have the market resources needed to engage in global value-creation activities, the cross-border platform's agility enables them to a realignment of any available resources and the exploitation of opportunities presented by changing global market conditions (Zahoor *et al.*, 2024). Accordingly, the benefits of AI adoption, coupled with their agile nature, can enable ESMEs to attain market orientation, marketing planning and new product development advantages suited to improving IMCs (Manis and Madhavaram, 2023).

Mainstream studies in the international business domain have extensively discussed the relevance of strategic agility and proposed its primary aspects for international performance and success (Ahmmed *et al.*, 2021; Demir *et al.*, 2021; Zahoor *et al.*, 2024). Still, more studies are needed to connect marketing to agile strategies, particularly in the context of ESMEs, distinguishing the different underlying capabilities that lead to international advantage (Shams *et al.*, 2021). More importantly, despite the emergence of AI-enabled marketing capabilities (Manis and Madhavaram, 2023), limited attention has been paid to what impact combining AI adoption and strategic agility may have on IMCs. Therefore, it is expected that AI adoption, when coupled with strategic agility, will enhance IMCs, we hypothesize:

H3. Coupling AI adoption with strategic agility has a significant positive effect on IMCs.

In addition to the direct effects, we anticipate that IMCs mediate the effects of coupling AI adoption and strategic agility on product innovation and brand upgrading advantage in our proposed conceptual model. An enterprise that effectively integrates its IMCs is better positioned to leverage the AI and strategic agility of cross-border platforms to enhance creativity in new products and brand development. It is more likely to generate added value in GVCs because it can fulfill marketing-related needs. IMCs facilitate customer knowledge acquisition, price adjustments, product and brand development and the strategic adaptation of marketing tactics, enabling ESMEs to target foreign customers with differentiated offerings (Zahoor and Lew, 2023).

Product innovation is mainly reflected in tangible goods R&D and production and service innovation and process (Cui and Wu, 2017). Product innovation decisions typically involve utilizing AI-enabled marketing capabilities to determine the optimal product attributes for development and employing test markets to decide whether and to what degree the product will be accepted (Huang and Rust, 2021). For instance, AI can assist ESMEs in analysing foreign user feedback from social and e-commerce platforms to identify new product features or design directions; AI can also support ESMEs in simulation testing, thereby reducing R&D costs. One especially important research area concerns how well AI-driven algorithms forecast demand for really new markets, develop innovative products and expedite the overall product development process (Davenport *et al.*, 2020). Furthermore, the strategic flexibility offered by platforms allows ESMEs to adapt more effectively to AI-driven predictions (Shams *et al.*, 2021). For example, flexible global supply chain models enable ESMEs to adjust production plans based on market feedback swiftly, accelerating new product iteration.

Moreover, the absence of brand effects not only diminishes the added value of products but also constrains the long-term international competitiveness of ESMEs. Building strong brands requires ESMEs to employ positioning and marketing mix strategies to shape foreign customers' perceptions (Pfajfar *et al.*, 2024b). AI-driven insights can tailor advertisements for foreign markets by classifying customers and evaluating potential advertisements based on their effectiveness in eliciting desired responses, thereby enhancing brand influence (Manis and Madhavaram, 2023). Besides, coupling AI adoption and strategic agility provides a foundation for IMCs to achieve effective allocation of advertising resources and manage cross-border customer relationships, thereby mitigating the constraints imposed by ESMEs' limited investments in marketing and brand building. IMCs facilitate brand differentiation, drive innovative branding strategies and align with international customers' preferences, thereby enhancing brand image and perceived value (Zahoor *et al.*, 2024). This enhanced positioning elevates ESMEs in GVCs by fostering customer loyalty and enabling premium pricing (Eng and Spickett-Jones, 2009). Thus, we propose the following hypotheses:

H4. IMCs positively mediate the effects of coupling AI adoption and strategic agility on (a) product innovation advantage and (b) brand upgrading advantage.

3.3 Moderating effects of ESMEs' platform embedding characteristics

We refer to embedding breadth as the number of digital platforms in which ESMEs participate, encompassing various types, such as e-commerce, social and innovation platforms (Kobarg

et al., 2019). Below, we will discuss how embedding breadth is associated with the effects of coupling AI adoption and strategic agility on ESMEs' international advantage, specifically in terms of product innovation and brand upgrading.

More specifically, when embedding breadth shifts from low to medium, ESMEs can access a wider variety of data through diverse platforms, enhancing their potential for creative recombination, complementary alignment within digital ecosystems and differentiated platform positioning, ultimately facilitating product innovation and brand upgrading (Hurtado-Torres *et al.*, 2018). In such conditions, the coupling of AI adoption and strategic agility enables the integration and analysis of diverse international market information, resulting in knowledge spillovers and unique product innovation advantages derived from heterogeneous resources, distributed innovation and cooperation among participants (Li *et al.*, 2019; Hagedoorn *et al.*, 2018). For example, interaction and reach platforms mostly comprise social media, facilitating product exchanges and providing access to essential data on prices, location and reputation. Additionally, different platforms allow ESMEs to tailor branding strategies based on platform positioning and target demographics, ultimately advancing brand equity (Liu *et al.*, 2025). For example, AI-driven rankings and review scores on e-commerce platforms can communicate product quality to foreign consumers, while social platforms facilitate interactions that build brand trust (Cutolo and Kenney, 2021). Consequently, increasing embedding breadth amplifies the effects of coupling AI adoption and strategic agility on ESMEs' international advantages.

By contrast, when embedding breadth shifts from medium to high, ESMEs may encounter a "learning trap", increase in information processing, resource dispersion and attention allocation challenges (Hagedoorn *et al.*, 2018; Kobarg *et al.*, 2019). As platform types increase, the degree of complexity of the portfolio and the difficulty of information processing, coordination and integration across platforms also escalate (Mariani and Nambisan, 2021; Li *et al.*, 2022). In such conditions, even with the adoption of AI and high strategic agility, the limited attention ESMEs can allocate across multiple platforms may restrict resources for innovation activities (Kobarg *et al.*, 2019). Additionally, various platforms can lead to inconsistent brand messaging, hindering consumers' ability to form a clear brand identity and negatively impacting brand perception (Hagedoorn *et al.*, 2018). For example, Li *et al.* (2022) discover that multinational enterprises with a more diversified digital platform attention scope experience constrained international sales. In short, initially, ESMEs benefit from embedded variety in digital platforms; however, excessive diversification can diminish these advantages due to rising coordination costs and declining information quality (Hurtado-Torres *et al.*, 2018; Ko *et al.*, 2021). Combining the discussions above, we formally propose that:

- H5. The breadth of platform embedding has an inverted U-shaped moderating effect on the positive relationship between coupling AI adoption and strategic agility and (a) product innovation advantage and (b) brand upgrading advantage.

Embedding depth reflects the resource investment ESMEs dedicate to cross-border digital platform activities and the intensity of their interactions with platform owners and participants (Li *et al.*, 2019; Kobarg *et al.*, 2019). We also anticipate that embedding depth moderates the effect of coupling AI adoption and strategic agility on international advantage.

More precisely, as embedding depth shifts from low to medium, ESMEs invest targeted resources and relationships into digital platforms, fostering close collaboration and tacit knowledge transfer (Li *et al.*, 2019). Dahlke *et al.* (2024) argue that the effectiveness of direct transmission of AI knowledge relies on the strength of links and the depth of shared knowledge among partners. By deeply engaging with digital platforms, ESMEs can establish AI-based organisational learning routines and extract AI-driven international marketing insights (Manis and Madhavaram, 2023). This engagement promotes resource transfer and integration, allowing ESMEs to leverage platform agility to shorten R&D cycles (Hagedoorn *et al.*, 2018). In addition, AI-powered, strategically agile organisations establish access mechanisms and bolster brand credibility. ESMEs grow alongside these platforms, resulting in a bundling of

their brand images. The global influence of large digital platforms enhances ESMEs' international visibility, particularly when consumers are cautious about new brands, as platform credibility fosters trust (Dahlke *et al.*, 2024). Furthermore, frequent interactions with global consumers also sustain a strong brand community atmosphere (Spyropoulou *et al.*, 2011). Hence, increased embedding depth positively moderates the effect of coupling AI adoption and strategic agility on product innovation and brand upgrading.

However, this positive moderating effect may diminish as embedding depth rises. A higher embedding depth is similar to a fairly closed system for AI adopters that may limit ESMEs' abilities to value and process distant and novel information (Hagedoorn *et al.*, 2018). This may lead to a "familiarity trap", where knowledge integration restricts the flow of new insights and reduces the novelty of the platform's knowledge base (Cutolo and Kenney, 2021). Embedding depth can also result in relational inertia, making ESMEs rigid in their platform type preferences. Over-dependence on specific platforms restricts strategic flexibility and hinders resource reallocation for relevant product innovation (Hagedoorn *et al.*, 2018). In addition, ESMEs employing a single-platform strategy may encounter power imbalances, as platform control limits their ability to implement independent branding strategies. Furthermore, a lack of marketing channel diversification can impede effective brand promotion (Cutolo and Kenney, 2021). Considering the discussion above, we postulate that varying levels of platform embedding depth affect ESMEs' ability to develop new products and upgrade brands in the global markets:

- H6.* The depth of platform embedding has an inverted U-shaped moderating effect on the positive relationship between coupling AI adoption and strategic agility and (a) product innovation advantage and (b) brand upgrading advantage.

This study proposes the research model presented in Figure 1, which demonstrates the hypothesized relationships between the coupling of AI adoption and strategic agility, IMCs, platform embedding breadth and depth, product innovation advantage and brand upgrading advantage.

4. Research methods

4.1 Study setting and data

To test the hypotheses, this study employed a questionnaire-based survey to explore how Chinese SMEs build international competitive advantage through cross-border platforms,

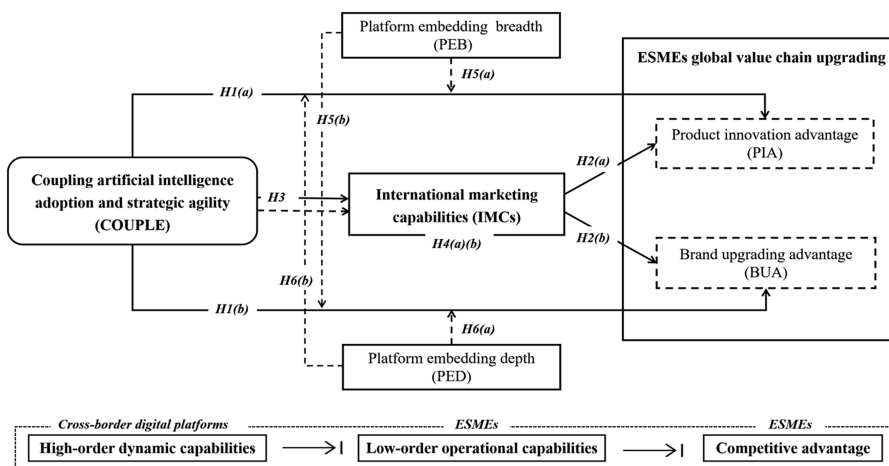


Figure 1. Research model. Source(s): Authors' own work

focusing on their embedding of B2B and B2C platforms. The sample data were collected from Chinese SMEs for the following reasons, justifying our selection as the research background. *First*, China is a key player in the GVC, with a vast number of export SMEs and cross-border platforms. By 2024, China had over 120,000 cross-border e-commerce firms and the number of independent cross-border e-commerce sites had surpassed 200,000, marking a significant advancement in the internationalization of SMEs through cross-border platforms. *Second*, Chinese SMEs are advancing from the middle and low to higher ends of the GVC, enhancing the added value of export products to meet the global demand for personalized, high-quality and branded goods. For example, Alibaba International Station has introduced a range of AI tools for SMEs, including text translation, keyword extraction, product page generation and market analysis. These tools create new opportunities for product innovation and brand upgrading of SMEs. Currently, over 30,000 SMEs globally use Alibaba's AI business assistant. Therefore, the pursuit of upgrading the GVC by Chinese SMEs through applying cross-border platforms' AI and strategic agility provides an ideal context to test our research model.

To ensure accuracy and cultural appropriateness, the survey was translated into Chinese using standard translation and back-translation procedures (Brislin, 1980). Moreover, before conducting the actual survey and following discussions on the proposed questionnaire, we pre-tested it with five business professionals to ensure the questions were clear, easily answerable and relevant to the industry. Their feedback led to revisions that improved clarity and finalized the questionnaire design.

After finalizing the questionnaire, we engaged a third-party independent survey agency from the Wen-Juan-Xing database, one of China's largest data collection platforms, which has over 620,000 registered managers and has completed 21.9 billion surveys by 2024. We selected SMEs in China, where classification is based on companies employing fewer than 2,000 staff, typically defined as SMEs (Tian *et al.*, 2018). In the contract, we clearly stated our data collection requirements, including the definition of SMEs, the random sampling of SMEs from different industrial sectors, and the inclusion of multinational companies, foreign trade firms and businesses engaged in cross-border operations. In addition, our respondents had extensive experience in business analytics and operational management, with the ability to apply these skills across various business functions.

They met three key criteria: (1) employment in foreign trade firms or departments engaged in international business; (2) involvement with cross-border digital platform-related activities; and (3) holding a managerial position. This process led to the identification of 6,600 eligible managers, to whom the online link to the final questionnaire was emailed for participation in the survey to test our hypotheses. Meanwhile, the questionnaire included a dummy variable for participation in cross-border platforms and a multiple-choice question regarding the specific platforms used for international business. The platforms listed included Alibaba International Station, Global Express, Amazon, TikTok, Temu, SHEIN and others. We asked SME managers to choose at least one cross-border platform through which their firms had internationalized their activities and to answer the questions from this perspective.

Ultimately, a total of 949 questionnaires were collected, yielding a 14.38% response rate, with 322 valid responses, representing a 33.94% effective response rate. The established criteria confirmed that the sample of Chinese SMEs were actively utilizing digital platforms to gain a competitive edge in the GVC. Anonymity was fully maintained throughout the data collection process. Table 1 summarizes the sample demographics, showing that a significant majority of the respondents worked in private companies (74%) and most belonged to manufacturing industries (64%). Most of the participants were between 30 and 39 years old, and they had more than five years of work experience in international business. In the context of the internationalization of sample companies through cross-border digital platforms, Alibaba International Station and Amazon are the two major cross-border platforms that enterprises join most, accounting for 78.9% and 71.4%, respectively, 33.9% of the companies chose to cooperate with AliExpress. Of course, the data clearly shows that more enterprises not only embed one platform but also combine several platforms. All sample respondents are

Table 1. Sample demographics ($n = 322$)

Demographic	Category	Frequency ($n = 322$)	Percentage (%)
Gender	Male	115	35.7%
	Female	207	64.3%
Age	<29	30	9.3%
	30–39	248	77.0%
	40–49	40	12.4%
	50–59	4	1.3%
Educational background	Doctor	6	1.9%
	Master	67	20.8%
	Undergraduate	241	74.8%
	Undergraduate degree below	8	2.5%
Position	Low-level managers	12	3.8%
	Mid-level managers	240	74.5%
	Top managers	70	21.7%
Firm size	<100	22	6.8%
	100–300	129	40.1%
	300–500	68	21.1%
	500–1,000	61	18.9%
	>1,000	42	13.1%
Industrial	Manufacturing	207	64.3%
	High-tech	34	10.6%
	Internet	34	10.6%
	Medical technology	4	1.2%
	New materials	10	3.1%
	Electronic	12	3.7%
	Clothing industry	10	3.1%
	Beauty industry	7	2.2%
Tenure	Household industry	4	1.2%
	<5	49	15.2%
	5–10	217	67.4%
	10–20	51	15.8%
	>20	5	1.6%
Ownership	State-owned	23	7.2%
	Private	239	74.2%
	Mixed-ownership	38	11.8%
	Foreign-funded	22	6.8%
Platform type	Alibaba International Station	254	78.9%
	AliExpress	109	33.9%
	Amazon	230	71.4%
	TikTok	99	30.7%
	Temu	88	27.3%
	SHEIN	15	4.7%

Source(s): Authors' own work

aware of the benefits of internationalization by embedding cross-border platforms and ongoing product and brand pilot initiatives around the world.

4.2 Variables and measurement

In this study, all measurement instruments for the research model constructs were adapted from prior studies. A seven-point Likert scale (1 = strongly disagree, 7 = strongly agree) was employed for all items.

Dependent variables. The coupling of AI adoption and strategic agility is conceptualized and developed as a formative–formative type of second-order construct, reflected by the AI

adoption and strategic agility first-order construct, which was previously validated by [Ameen et al. \(2024\)](#). AI adoption items were taken from [Hossain et al. \(2022\)](#), which evaluate the extent to which digital platforms actively utilize AI to interpret international data correctly and derive insights. Following [Hossain et al. \(2022\)](#), we used a four-item scale to measure AI adoption on the platform, asking ESME managers how AI helps design products and services, manage and optimize supply chains and reduce shipping costs and transit time, accurate predictions of inventory demand and accelerates the strategic decision process. Strategic agility was measured through eight statements using a framework adapted from [Tallon and Pinsonneault \(2011\)](#), which emphasizes the speed and flexibility in leveraging AI-driven learnings to achieve specific goals and tasks on the platform. The items for strategic agility, such as respond to changes in consumer demand, customize products to suit an individual customer, introduce new product or pricing schedules in response to changes in competitors, expand into new international markets and adopt new technologies to produce better (see [Table 4](#)).

Mediating variable. We employed a five-item IMCs scale, adapted from [Yang et al. \(2019\)](#) and validated by [Zahoor and Lew \(2023\)](#), which covers product modification according to the export market, switching from the current export market, modifying existing products and resources to meet export market needs and switching the uses and applications of resources.

Dependent variables. While product innovation and brand upgrading advantages were measured using scales from [Cui and Wu \(2017\)](#) and [Spyropoulou et al. \(2011\)](#), respectively. The product innovation advantage was evaluated using four indicators of innovativeness, specifically the degree to which a new product is novel to the industry and introduces original concepts. And the brand upgrading advantage was assessed using a four-item scale to evaluate the brand position of an ESMEs' offering in the export market, focusing on brand image, awareness, "share of mind" and personality, relative to its major competitor.

Moderating variables. Platform embedding characteristics were divided into two dimensions: breadth, which refers to the extent to which firms engage with multiple digital platforms and depth, reflecting the intensity of collaboration with these platforms ([Kobarg et al., 2019](#); [Mariani and Nambisan, 2021](#)). To assess platform embedding breadth and depth, we modified scales from [Xiao et al. \(2020\)](#), originally designed for international networks of ESMEs and combined them with measures from [Kobarg et al. \(2019\)](#) to evaluate managers' perceptions of their company's platform embeddedness in terms of breadth and depth. Breadth and depth were calculated by the scores of the three and four items per construct, respectively. The key items are listed in [Table 4](#).

Control variables. Additionally, to mitigate potential biases and keep them in line with prior literature, we incorporated relevant control variables. Respondents provided information on both firm-level and individual-level characteristics. Specifically, firm size, firm age, ownership type, industry category, international sales and profit were included in the research model. Furthermore, we controlled for individual factors such as gender, age, education level, position and tenure.

5. Analysis and results

Data analysis was conducted using SPSS 27, AMOS 24 and STATA to evaluate both the measurement and structural models. SPSS 23 was employed for demographic analysis, descriptive statistics and reliability analyses. The measurement instruments were tested through confirmatory factor analysis (CFA) using AMOS 24, while STATA was utilized for the final moderation effect tests ([Gerbing and Anderson, 1988](#)). The analysis proceeded in two stages: assessing the measurement model's quality, followed by testing the structural model to evaluate the research hypotheses. Finally, to validate the original results, we also conducted rigorous additional and robustness analyses.

[Table 2](#) presents the descriptive analysis of the variables, including means, standard deviations and inter-variable correlations. The mean scores for all items across the constructs

Table 2. Correlation matrix, means and standard deviations

Variables	COUPLE	IMC	PIA	BUA	PEB	PED
COUPLE	1	0.302**	0.418**	0.309**	0.352**	0.343**
IMC	0.302**	1	0.601**	0.461**	0.320**	0.417**
PIA	0.418**	0.601**	1	0.444**	0.299**	0.431**
BUA	0.309**	0.461**	0.444**	1	0.617**	0.254**
PEB	0.352**	0.320**	0.299**	0.617**	1	0.376**
PED	0.343**	0.417**	0.431**	0.254**	0.376**	1
Mean	5.502	5.483	5.492	5.614	5.428	5.710
SD	0.789	1.010	0.991	0.846	1.188	0.920

Note(s): * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Source(s): Authors' own work

exceeded the neutral point (4), indicating that respondents generally agreed with the items. The results of the inter-variable correlations provide initial support for our hypotheses.

5.1 Measurement model

We assessed the reliability, content validity, convergent validity and discriminant validity of the measurement model using several statistical methods. For reliability testing, we used Cronbach's alpha and composite reliability (CR). The Cronbach's alpha values for the constructs ranged from 0.799 to 0.889, while the CR values varied from 0.801 to 0.890 (see Table 4). All the metrics exceeded the recommended threshold of 0.7, indicating good reliability (Hair *et al.*, 2010). Next, we examined the factor loadings, which, ideally, should be greater than 0.7, with a minimum acceptable threshold of 0.6 (Fornell and Larcker, 1981). The factor loadings for the remaining items ranged from 0.673 to 0.876, with all exceeding the minimum threshold of 0.65.

CFA was validated in two stages: (1) assessing goodness-of-fit indices and (2) assessing construct validity (Hair *et al.*, 2010). The results revealed that the data fit the model well ($\chi^2/\text{df} = 1.361$; goodness of fit index (GFI) = 0.900; comparative fit index (CFI) = 0.967; Tucker–Lewis index (TLI) = 0.964; incremental fit index (IFI) = 0.968; root mean square error of approximation (RMSEA) = 0.034). The χ^2/df exceeded the minimum requirement of 1:3, while the values for GFI, IFI, TLI and CFI all surpassed the recommended threshold of ≥ 0.90 . RMSEA also meet the criteria of < 0.05 (see Table 3).

In the second stage, we validated content validity, convergent and discriminant validity. First, the measurement items are all derived from mature scales published in authoritative journals and have been verified, thus ensuring the measurement tool's content validity. Second, the average variance extracted (AVE) values for all constructs ranged from 0.503 to 0.708, surpassing the 0.50 threshold, which indicates adequate convergent validity (see Table 4). Third, to assess discriminant validity, we utilized the heterotrait-monotrait (HTMT) ratio method. We found that the square roots of the AVEs (shown on the diagonal of the matrix) were greater than the correlations between constructs, thereby confirming discriminant validity (Fornell and Larcker, 1981) (see Table 5).

5.2 Structural model

We employed covariance-based structural equation modelling (CB-SEM) to rigorously test our hypotheses. This method is particularly suitable for theory testing, especially when working with relatively simple models and sufficient sample sizes (Hair *et al.*, 2010). To examine mediation effects, we applied bootstrapped regression analysis following Hayes' (2013) methodology. Additionally, we used STATA to conduct further regression analyses to assess the nonlinear moderating effects of platform embedding characteristics. The results are illustrated in Figure 3 and Table 8.

Table 3. Confirmatory factor analysis

Model	Factors	χ^2/df	GFI	CFI	TLI	IFI	RMSEA
Model 1	Six factors: COUPLE; IMCs; PIA; BUA; PEB; PED	1.361	0.900	0.967	0.964	0.968	0.034
Model 2	Five factors: COUPLE; IMCs; PIA + BUA; PEB; PED	2.069	0.824	0.902	0.893	0.903	0.058
Model 3	Four factors: COUPLE; IMCs; PIA + BUA; PEB + PED	2.877	0.734	0.827	0.811	0.828	0.076
Model 4	One factor: COUPLE + IMCs + PIA + BUA + PEB + PED	6.151	0.127	0.571	0.516	0.483	0.519
Source(s): Authors' own work							

Table 4. Measurement items

Constructs and items	Loadings	Cronbach's alpha	CR	AVE
<i>Artificial Intelligence Adoption (AI)</i> (1 = strongly disagree, 7 = strongly agree)				
AI1: AI adoption help firm to design product and services distinctly	0.771	0.815	0.818	0.529
AI2: AI can manage and optimize supply chains and reduce shipping costs and transit time	0.674			
AI3: AI makes more accurate predictions of inventory demand and therefore reduces wastage	0.740			
AI4: AI adoption accelerates the strategic decision process by measuring marketing predictions accurately	0.721			
<i>Strategic Agility (SA)</i> (1 = strongly disagree, 7 = strongly agree)				
How easily and quickly can your firm perform the following actions?				
SA1: Respond to changes in aggregate consumer demand	0.771	0.889	0.890	0.503
SA2: Customize a product to suit an individual customer	0.688			
SA3: React to new product launches by competitors	0.707			
SA4: Introduce new pricing schedules in response to changes in competitors' prices	0.700			
SA5: Expand into new regional or international markets	0.688			
SA6: Change (i.e. expand or reduce) the variety of products available for sale	0.730			
SA7: Adopt new technologies to produce better, faster and cheaper products and services	0.711			
SA8: Switch suppliers to avail of lower costs, better quality or improved delivery times	0.673			
<i>International Marketing Capabilities (IMC)</i> (1 = strongly disagree, 7 = strongly agree)				
IMC1: Adjust the product line according to the export market	0.803	0.869	0.870	0.572
IMC2: Switch from current export market to different target	0.694			
IMC3: Modify existing products to meet export market needs	0.736			
IMC4: Adjust how resources are used to meet export market needs	0.761			
IMC5: Switch the uses and applications of resources to meet export market needs	0.784			
<i>Product Innovation Advantage (PIA)</i> (1 = strongly disagree, 7 = strongly agree)				
The new product developed from this project				
PIA1: Was of higher quality than competing products	0.792	0.852	0.854	0.594
PIA2: Was superior to competing products in terms of meeting customers' needs	0.746			
PIA3: Offered unique benefits to the customers	0.754			
PIA4: Performed better than competitors' products	0.791			
<i>Brand Upgrading Advantage (BUA)</i> (1 = strongly disagree, 7 = strongly agree)				
Export venture branding advantage				
BUA1: Was of better image than competing brands	0.751	0.799	0.801	0.503
BUA2: Was of higher awareness than competing brands	0.691			
BUA3: Offered unique share of mind to the customers	0.663			
BUA4: Performed more personality than competitors' products	0.728			

(continued)

Table 4. Continued

Constructs and items	Loadings	Cronbach's alpha	CR	AVE
<i>Platform Embedding Breadth (PEB)</i> (1 = strongly disagree, 7 = strongly agree)				
PEB1: Compared with our competitors, our company has cooperated with multiple cross-border digital platforms	0.828	0.877	0.879	0.708
PEB2: Our company constantly spends substantial time and effort in building networks with different participants on the platform, including suppliers, foreign customers, competitors, government agencies, consultants, R&D companies and research institutions	0.819			
PEB3: Our company searches for innovative ideas from different channels around the world through the platform to obtain knowledge	0.876			
<i>Platform Embedding Depth (PED)</i> (1 = strongly disagree, 7 = strongly agree)				
PED1: The international network of the cooperation platform provides important resources for the internationalization of our company	0.793	0.831	0.833	0.555
PED2: Our company has considerable commitments to important partner platforms and users	0.679			
PED3: Our company has many human resources for cross-border platform operations	0.742			
PED4: Our company promises to frequently contact with embedded cross-border platforms	0.762			
Source(s): Authors' own work				

Table 5. Discriminant validity

Construct	1	2	3	4	5	6
Coupling of AI adoption and Strategic Agility	0.797					
International Marketing Capabilities	0.302	0.756				
Product Innovation Advantage	0.418	0.601	0.771			
Brand Upgrading Advantage	0.309	0.461	0.444	0.709		
Platform Embedding Breadth	0.352	0.320	0.299	0.617	0.841	
Platform Embedding Depth	0.343	0.417	0.431	0.254	0.376	0.745
Source(s): Authors' own work						

Direct effects analysis. Our test showed that the coupling of AI adoption and strategic agility exerted significant positive effects on both product innovation advantage (H1a: $\beta = 0.519$, $p = < 0.001$) and brand upgrading advantage (H1b: $\beta = 0.336$, $p = < 0.05$), thus supporting H1a and H1b.

The analysis confirms that IMCs are significantly associated with product innovation advantage (H2a: $\beta = 0.498$, $p < 0.001$) and brand upgrading advantage (H2b: $\beta = 0.367$, $p < 0.001$), thus supporting H2a and H2b. Furthermore, the coupling of AI adoption and strategic agility has a significant positive impact on IMCs (H3: $\beta = 0.696$, $p < 0.001$), providing support for H3 (see Table 6).

Mediation effects analysis. As shown in Table 7, we employed the bootstrap method with 5,000 samples and a 95% confidence interval (CI) for mediated models. The results indicate that an ESME's IMCs mediate the relationship between the coupling of AI adoption and strategic agility on product innovation advantage ($\beta = 0.346$; CI = [0.197, 0.566]) and brand upgrading advantage ($\beta = 0.242$; CI = [0.117, 0.467]). The 95% CI's upper and lower bounds

Table 6. Path hypothesis testing

Structural paths	Estimate	S.E.	C.R.	p-value	Hypothesis test
COUPLE → PIA	0.519	0.109	4.764	***	H1a: Supported
COUPLE → BUA	0.336	0.106	3.168	**	H1b: Supported
IMC → PIA	0.498	0.056	8.901	***	H2a: Supported
IMC → BUA	0.367	0.056	6.589	***	H2b: Supported
COUPLE → IMCs	0.696	0.144	4.845	***	H3: Supported

Source(s): Authors' own work

Table 7. Bootstrapped mediation results

Path	Coefficient Estimate	SE	Bias-corrected 95% CI Lower	Upper
H4a.COUPLE → IMC → PIA	0.346	0.092	0.197	0.566
H4b.COUPLE → IMC → BUA	0.255	0.087	0.117	0.467

Note(s): Results obtained using 5,000 bootstrap samples. SE = standard error, CI = confidence interval

Source(s): Authors' own work

do not include zero, indicating that the mediation effects for both H4a and H4b are supported. The results of path coefficient estimates are illustrated in Figure 2.

Moderated effects analysis. To test the moderating effects of platform embedding characteristics on the relationship between the coupling of AI adoption and strategic agility with international advantage (i.e. H5–H6), we employed a stepwise regression analysis. Table 8 presents the results of our regression analysis. Following the recommendations of Aiken and West (1991), we tested for nonlinear moderating effects using the equation provided, where three antecedent variables – X, Z and Z² (the quadratic term of Z), XZ (the product of X and Z) and XZ² (the product of X and Z²) – predict the dependent variable Y. Here, α represents the intercept, β₁ ~ β₅ are the regression coefficients for X, Z, Z², XZ and XZ² and ε is a residual term. The product term (XZ²) allows us to test for the nonlinear moderating effect of Z on the relationship between X and Y.

$$Y = \alpha + \beta_1 X + \beta_2 Z + \beta_3 Z^2 + \beta_4 XZ + \beta_5 XZ^2 + \varepsilon$$

We applied this approach to examine the nonlinear moderating effects of platform embedding breadth (PEB) on the relationship between the coupling of AI adoption and strategic agility (COUPLE) with product innovation advantage (PIA) and brand upgrading advantage (BUA). Similarly, we tested the nonlinear moderating effects of platform embedding depth (PED) on the relationship between COUPLE and both PIA and BUA. The specific equations are outlined as follows:

- (1) $PIA = \alpha + \beta_1 COUPLE + \beta_2 PEB + \beta_3 PEB^2 + \beta_4 COUPLE \times PEB + \beta_5 COUPLE \times PEB^2 + \varepsilon$
- (2) $PIA = \alpha + \beta_1 COUPLE + \beta_2 PED + \beta_3 PED^2 + \beta_4 COUPLE \times PED + \beta_5 COUPLE \times PED^2 + \varepsilon$
- (3) $BUA = \alpha + \beta_1 COUPLE + \beta_2 PEB + \beta_3 PEB^2 + \beta_4 COUPLE \times PEB + \beta_5 COUPLE \times PEB^2 + \varepsilon$
- (4) $BUA = \alpha + \beta_1 COUPLE + \beta_2 PED + \beta_3 PED^2 + \beta_4 COUPLE \times PED + \beta_5 COUPLE \times PED^2 + \varepsilon$

Table 8. Results of moderating effects

	Product innovation advantage			Brand upgrading advantage		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
<i>Controls</i>						
Gender	−0.027 (0.104)	−0.015 (0.100)	0.060 (0.090)	−0.032 (0.094)	−0.022 (0.069)	−0.007 (0.093)
Age	−0.029 (0.099)	−0.085 (0.096)	−0.035 (0.085)	0.087 (0.089)	−0.018 (0.066)	0.090 (0.088)
Education level	0.095 (0.097)	0.093 (0.094)	0.097 (0.083)	−0.010 (0.087)	0.019 (0.065)	−0.004 (0.086)
Position	0.047 (0.101)	0.054 (0.098)	−0.048 (0.087)	0.143 (0.091)	0.094 (0.068)	0.110 (0.090)
Size	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Year	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
International sales	0.158 (0.066)**	0.142 (0.064)**	0.075 (0.057)	0.092 (0.059)	0.032 (0.044)	0.055 (0.059)
International profit	0.140 (0.067)**	0.133 (0.065)**	0.134 (0.057)	0.036 (0.061)	0.037 (0.045)	0.035 (0.059)
Industry category	0.040 (0.024)	0.043 (0.024)	0.037 (0.021)	−0.004 (0.022)	−0.011 (0.016)	−0.004 (0.022)
Tenure	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)**	0.000 (0.000)	0.000 (0.000)
Ownership type	0.005 (0.075)	0.011 (0.074)	0.006 (0.064)	−0.180 (0.068)	−0.102 (0.051)	−0.181 (0.067)*
<i>Independent Variables</i>						
COUPLE	0.540 (0.072)***	0.507 (0.078)***	0.520 (0.068)***	0.352 (0.065)***	0.202 (0.053)***	0.331 (0.070)***
<i>Moderator</i>						
PEB		0.259 (0.075)***			0.530 (0.052)***	
PEB ²		0.083 (0.041)**			0.060 (0.028)	
PED			0.438 (0.076)***			0.226 (0.079)**
PED ²			0.008 (0.026)			0.028 (0.027)
<i>Interactions</i>						
COUPLE × PEB		0.144 (0.076)*			0.029 (0.052)	
COUPLE × PEB ²		−0.012 (0.077)			−0.233 (0.053)***	
COUPLE × PED			0.075 (0.062)			0.011 (0.065)
COUPLE × PED ²			−0.216 (0.063)***			−0.093 (0.065)
<i>Model Summary</i>						
Constant	0.553 (0.629)	−0.567 (0.645)	−1.293 (0.629)	2.863 (0.567)***	1.210 (0.444)*	1.891 (0.652)**
F-value	8.51	8.68	16.43	5.010	23.600	5.040
p-value	0	0	0	0.000	0.000	0.000
R-Square	0.2484	0.3128	0.4629	0.163	0.553	0.209
Adjusted R-Square	0.2192	0.2768	0.4347	0.130	0.530	0.168
Source(s): Authors' own work						

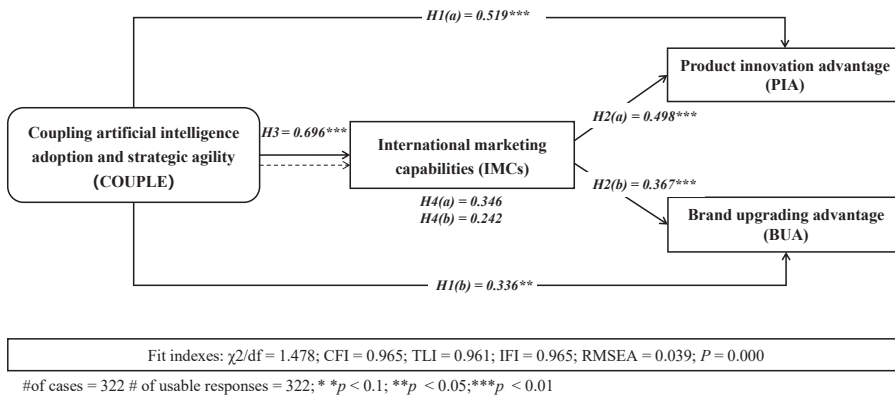


Figure 2. Structural model with path coefficient estimates. **Source(s):** Authors' own work

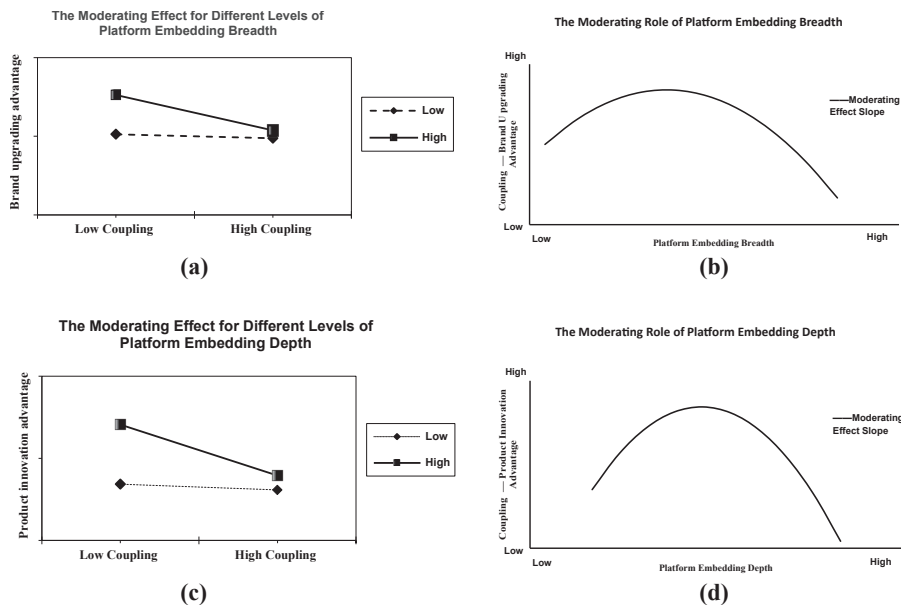


Figure 3. Graphical representation. **Source(s):** Authors' own work

Hypothesis 5a suggests that platform embedding breadth has an inverted U-shaped moderating effect on the relationship between the coupling of AI adoption with strategic agility and product innovation advantage. However, Model 2 indicates that the coefficient of the interaction effect between COUPLE and the square of PEB is non-significant ($\beta = -0.012$, $p = 0.874 > 0.05$). Therefore, **H5a** is not supported.

Hypothesis 5b posits that platform embedding breadth has an inverted U-shaped moderating effect on the relationship between coupling AI adoption with strategic agility and brand upgrading advantage. Model 5 shows that the interaction effect between COUPLE and the square of PEB is significant ($\beta = -0.233$, $p = 0.000 < 0.001$). Therefore, **H5b** is supported.

Hypothesis 6a posits that platform embedding depth exerts an inverted U-shaped moderating effect on the relationship between the coupling of AI adoption with strategic agility and product innovation advantage. The results from Model 3 show that the interaction of COUPLE and the square of PED is significant ($\beta = -0.216, p = 0.001$), supporting **H6a**.

In **Hypothesis 6b**, we hypothesise that platform embedding depth has an inverted U-shaped moderating effect on the relationship between the coupling of AI adoption with strategic agility and brand upgrading advantage. However, contrary to our hypothesis, the results from Model 6 indicate that the interaction between COUPLE and the square of PED is non-significant ($\beta = -0.093, p = 0.151 > 0.05$). Therefore, **H6b** is not supported.

Furthermore, the interaction plots in **Figure 3a and 3b** demonstrate the nonlinear moderating effects of platform embedding breadth on the relationship between coupling and brand upgrading advantage. **Figure 3c and 3d** demonstrate the nonlinear moderating effect of platform embedding depth on the relationship between coupling and product innovation advantage. These findings support **Hypotheses H5b and H6a**.

5.3 Testing for response and non-response

We followed **Armstrong and Overton's (1977)** procedure to test for non-response bias by comparing the first and last 25% of responses across key variables. We conducted *t*-tests on diverse organizational characteristics, such as industry type and firm size, to compare the two groups. The *t*-test results indicated no statistically significant differences between the two subsamples in terms of industry type ($t = -0.362, p = 0.719$) and firm size ($t = -1.271, p = 0.208$), indicating that no systematic non-response bias was present for the data collected.

5.4 Common method bias

To address potential common method bias (CMB) from single-informant data, we implemented several preventive approaches. During data collection, we included redundant questions in the questionnaire to minimise simultaneity bias and the effects of common method variance (CMV) (**Chang et al., 2010**). Additionally, respondents were kept unaware of the study's objectives to reduce evaluation concerns and social desirability bias and maintain the integrity of their responses. To further assess CMB, we conducted Harman's single-factor test, which revealed that a single factor accounted for only 33.08% of the total variance – well below the 50% threshold (**Podsakoff et al., 2003**), indicating minimal CMB risk. Moreover, the results of the single latent factor analysis ($\chi^2/\text{df} = 6.151$; RMSEA = 0.127; GFI = 0.571; CFI = 0.516; TLI = 0.483; IFI = 0.519) differed from those of the six-factor confirmatory model ($\chi^2/\text{df} = 1.361$; RMSEA = 0.034; GFI = 0.900; CFI = 0.967; TLI = 0.964; IFI = 0.968).

We employ **Kock's (2015)** full collinearity test to complement the previously suggested collinearity test. According to **Kock and Lynn (2012)**, collinearity is typically indicated when the variance inflation factor (VIF) exceeds 3.3, suggesting the presence of CMV. In our analysis, all VIF values remain below this threshold ($\text{VIF} < 1.101$), confirming that any potential effects of CMV are negligible. In addition, to establish the validity and reliability of the higher-order construct of coupling of AI adoption and strategic agility, we first assessed the weights of the formative lower-order constructs on their corresponding higher-order constructs (comprising two first-order constructs). All weights were significant and positively associated with their assigned higher-order construct. Next, we examined potential multicollinearity among the first-order constructs. All VIF values for the first-order constructs were below the 3.3 threshold, indicating that multicollinearity was not a concern within our sample (see **Table 9**).

5.5 Robustness checks

We tested two alternative research models. Our first alternative model treated the coupling of AI adoption and strategic agility reflective constructs as second-order “reflective—formative”

Table 9. Higher-order construct validation

Construct	Measures	Weight	Significance	VIF
Coupling of AI adoption and strategic agility	AI adoption	0.229	$p < 0.001$	1.111
	Strategic agility	0.274	$p < 0.001$	1.111

Source(s): Authors' own work

constructs. AI adoption and strategic agility were modelled as first-order constructs, and both their main effects and mediation effects were tested. The second alternative model defined product innovation advantage and brand upgrading advantage as first-order constructs of GVCs upgrading, with all constructs represented as reflective indicators. The results of the alternative model analysis indicate that coupling of AI adoption and strategic agility has a significant impact on IMCs, product innovation advantage and brand upgrading advantage. Furthermore, the mediation effects of IMCs were found to be statistically significant, even when GVC upgrading was specified as a first-order reflective construct. The results supported the hypothesized relationships. The robustness checks demonstrate that, whether a first- or second-order model was employed, no statistically significant differences were found in the tests of the main effects and mediation effects. The results of the robustness tests are detailed in Table 10.

Table 10. Test of robustness

Relationship/ hypothesis	Alternative 1 (model1)	Alternative 1 (model2)	Alternative 2 (model3)	Hypothesis test	
COUPLE → PIA (H1a)	0.265*** (4.666) [0.097, 0.232]			H1a: Supported	
COUPLE → BUA (H1b)		0.189*** (3.199) [0.066, 0.199]		H1b: Supported	
COUPLE → IMCs (H3)	0.307*** (5.012) [0.192, 0.432]	0.304*** (4.942) [0.190, 0.429]		H3: Supported	
COUPLE → IMCs → PIA (H4a)	0.160*** (4.800) [0.097, 0.232]			H4a: Supported	
COUPLE → IMCs → BUA (H4b)		0.124*** (3.602) [0.066, 0.199]		H4b: Supported	
COUPLE → GVC (H1ab)			0.274*** (5.213) [0.105, 0.244]	H1ab: Supported	
IMCs → GVC (H2ab)			0.554*** (11.120) [0.454, 0.650]	H2ab: Supported	
COUPLE → IMCs → GVC (H4ab)			0.170*** (4.844) [0.105, 0.244]	H4ab: Supported	
Coefficient of determination	R ²	Adjusted R ²	R ² 1.000	Adjusted R ²	Adjusted R ²
COUPLE	1.000	1.000	1.000	1.000	1.000
PIA	0.427	0.424			
BUA			0.249	0.244	
IMCs	0.094	0.091	0.092	0.090	0.094
GVC					0.475
Note(s): * <i>p</i> < 0.05, ** <i>p</i> < 0.01, *** <i>p</i> < 0.001. t-values and bootstrapping 95% CI bias corrections are indicated within parentheses and square brackets, respectively. GVC. Alternative 1 = Alternative COUPLE. Alternative 2 = Alternative GVC					
Source(s): Authors' own work					

5.6 Additional test

We also performed multiple regression analyses using mediation (5,000 re-samples) (Baron and Kenny, 1986), using SPSS on the primary research model. The results show that the IMCs mediate the relationship between coupling and product innovation advantage (total effects: 0.522***, $p < 0.001$) and between coupling and brand upgrading advantage (total effects: 0.404**, $p < 0.001$). The effects of the control variable were found to be non-significant ($p > 0.05$).

6. Discussion

6.1 Discussion of results

Digital technologies have significantly influenced international business, and SMEs have increasingly leveraged digital platforms to expand internationally in recent years (Li et al., 2019). These platforms, often multi-sided or two-sided, connect foreign customers with independent SME providers. Beyond becoming important conduits for international transactions, cross-border platforms serve as vital marketing channels for resource-constrained ESMEs, enabling them to secure a favourable global position in both product and brand. Our study investigates the role of AI adoption and strategic agility of platforms in promoting the international competitive advantage of ESMEs. We found that the coupling of AI adoption and strategic agility allows ESMEs to identify target foreign international opportunities and access market-specific insights and resources for product innovation and brand upgrading. Moreover, our results indicate that IMCs mediate the relationship between this coupling and two key international advantages. ESMEs can develop their IMCs through digital platforms to fulfil foreign market customers' expressed and latent needs, thus promoting international competitiveness.

More importantly, the results highlight that platform embedding characteristics are crucial moderating factors. Specifically, embedding breadth has an inverted U-shaped moderating effect on the relationship between the coupling of AI with strategic agility and brand upgrading advantage. That is, embedding breadth improves the effect of coupling AI with strategic agility on brand upgrading up to an optimal point (at a medium level). We also find that embedding depth has an inverted U-shaped moderating effect on the relationship between the coupling of AI with strategic agility and product innovation advantage. In other words, the effect of embedding depth on product innovation advantage becomes stronger when the depth of enterprise embeddedness in the platform is from low to medium, while when the depth of platform embeddedness is from medium to high, the impact of coupling on brand upgrading advantages weakens. These findings imply that while network engagement is beneficial, excessive breadth or depth does not always yield superior performance.

6.2 Theoretical implications

Building on research examining internationalization through digital platforms, this study's objective was to try to understand if and under what conditions cross-border platforms can enhance ESMEs' GVC positioning. To address this question, we consider cross-border platforms from the standpoint of key resources and capabilities that offer ESMEs competitive advantages. This study makes four significant theoretical contributions.

First, our study is one of the first to explore the coupling of AI adoption and strategic agility within digital platform reshaping product innovation and brand upgrading, providing insights for ESMEs to overcome the "low-end lock-in" in GVCs and achieving global sustainable development goals. To advance the theoretical understanding of firms' internationalization from emerging markets through digital platforms, we present a second-order aggregate concept examining the coupled effects of AI adoption and strategic agility. Specifically, this study extends existing research on digitalization and SME internationalization, responding to calls for deeper exploration of the intersection between digital platforms and

internationalization (Kim and Cavusgil, 2020; Luo, 2021). While existing research has explored key internationalization outcomes arising from the use of digital platforms, such as cost-effective market entry, international speed, scope and performance (Coviello *et al.*, 2017; Li *et al.*, 2019; Gong *et al.*, 2024), it also highlights the risk of a “virtuality trap” for platform, raising concerns about the ESMEs’ sustainable competitive advantage (Da Rocha *et al.*, 2024). Our research further examines the higher-level impact of digital platforms on internationalization efforts by introducing the distinctive process of AI adoption coupled with strategic agility, which enables ESMEs despite lacking strategic assets like proprietary technologies, global brands and market power, to real-time sense international market shifts and creatively leverage resources from cross-border channels and cooperative networks to upgrade their value chain position in global competition.

Second, the study extends the strategic agility literature by examining how digital platforms utilize agility not just as a high-level dynamic capability, but also as a strategic response to AI-enabled predictive insights and decision-making. While existing research has emphasized internal processes and strategies that enable agility, such as decision-making flexibility, adaptive leadership, exploitation and exploration strategies (Ahammad *et al.*, 2021; Shams *et al.*, 2021), this study advances understanding by demonstrating how AI adoption necessitates the development of agile capabilities. AI-enabled platforms enhance ESME’s ability to sense environmental shifts, reconfigure resources and respond with agility – key tenets of the dynamic capability framework (Teece *et al.*, 1997, 2016). While Strategic agility, defined as a firm’s ability to rapidly adapt strategies in response to external changes, traditionally emphasizes the resultant impact on the firm’s international performance (Ahammad *et al.*, 2021). The study demonstrates how synthesizing vast data and identifying actionable signals enhances strategic foresight and responsiveness in rapidly evolving demand patterns and customer needs. This contribution clarifies how combining agility with AI enables ESMEs to create and capture value in global environments marked by uncertainty, institutional inefficiency and market volatility.

Third, following the dynamic capability hierarchy view, the findings evidence the existence of a robust mechanism of the IMCs in the relationships between the coupling of AI adoption and strategy and international competitive advantage (Zollo and Winter, 2002; Winter, 2003; Schulze and Brusoni, 2022). By providing empirical insight into higher-order dynamic capabilities that drive “regular” operational capabilities for GVC upgrading, this research offers a significant opportunity to advance the dynamic interaction relationship between digital platforms and firms (Eng and Spickett-Jones, 2009). Previous research has examined the antecedents of digital platform use in internationalization, focusing on internal factors such as firm resources and capabilities, as well as organizational factors, including marketing and international orientation (e.g. Li *et al.*, 2019; Da Rocha *et al.*, 2024; Gong *et al.*, 2024). We concentrate on IMCs as specific operational capabilities, such as global rebranding and new product innovation, instead of only one general IMC, where IMCs reflect the processes ESMEs learn and implement to handle strategic marketing changes.

Additionally, this study also responds to Pfajfar *et al.* (2024b), who advocate for the application of IMCs within intricate GVCs by highlighting the critical role of robust IMCs in enabling product innovation and brand upgrading within the context of emerging economies (Eng and Spickett-Jones, 2009). Existing IMC research has predominantly focused on developed economies (e.g. Zahoor and Lew, 2023); studies in emerging economies provide insights by exploring the direct relationship between focal capabilities and company adaptiveness, export performance and financial performance (Pfajfar *et al.*, 2024a). Our findings extend AI-driven marketing capabilities to international business research, particularly when a company enters advanced internationalization but still has not attained global corporate status.

Finally, our study examines the nonlinear moderating role of platform embedding breadth and depth in the direct relationship between coupling AI adoption and strategic agility and two distinct outcomes: product innovation advantage and brand upgrading advantage. While

existing research has pointed out that the boundary conditions of internationalization were mostly focused on macro-level contingencies, such as institutional pressures, foreign market uncertainty and country-specific environmental factors (e.g. [Adomako et al., 2019](#); [Ko et al., 2021](#)). However, little is known about how platform embedding characteristics shape the effectiveness of AI adoption and strategic agility. This nuanced understanding provides a more refined model for theoretical predictions and enriches the academic discourse on the embedding breadth and depth and its role in shaping ESMEs' product innovation and brand upgrading advantage.

We observed embedding depth has an inverted U-shaped moderating effect on the positive relationship between coupling AI adoption and strategic agility and product innovation advantage; however, it does not moderate the relationship between coupling AI adoption with strategic agility and brand upgrading. The findings suggest that greater depth in platform embeddedness facilitates the transfer of tacit knowledge generated by AI, thus reducing the risk of product innovation and enhancing the exploratory level for innovation. However, over-reliance on specific platforms limits access to diverse market information, creating a 'familiarity trap' that hinders the reallocation of resources necessary for product innovation. This pattern is consistent with [Kobarg et al.'s \(2019\)](#) observation of a similar inverted U-shaped relationship between collaboration depth and radical innovation. Similarly, the impact of embedding breadth on brand upgrading follows an inverted U-shape. ESMEs embedded across multiple platforms enable SMEs to harness platform-specific differences, tailor brand strategies and deliver targeted brand messages underpinned by platform positioning and user preferences, driving overall brand exposure. However, excessive embedding risks a "learning trap", where limited attention allocation hinders the integration of fragmented multi-platform information, potentially leading to inconsistent brand messages received by users, thus damaging the brand image. This finding extends the research of [Xiao et al. \(2020\)](#), who highlighted the significance of international network breadth in enhancing global performance. Our findings cautioned that an excessive network scope may dilute brand focus, potentially undermining brand equity.

6.3 Practical implications

The findings of this study carry several crucial implications for managers striving to GVC upgrade via digital platforms. *First*, owners of cross-border platforms should recognize AI adoption and strategic agility as critical drivers of participants' product innovation and brand upgrading. AI adoption significantly enhances the platform's potential for identifying international market opportunities by collecting, integrating, analyzing and leveraging user and international market knowledge. This enables efficient and effective redeployment of their resources to value-creating and value-capturing higher-yield activities as AI-driven insight warrants. However, AI adoption can be costly and needs to be married to a sound strategy and agility. AI provides direction and goals for agility, while agility supports the effective implementation of AI-driven insight. Without considering such a coupling relationship, the value and necessity of AI adoption cannot be properly presented. Therefore, incorporating strategic agility throughout AI adoption processes is crucial, ensuring a smooth creation and protecting the value of new digital tools.

Second, IMC is a key mechanism through which the coupling of AI adoption and strategic agility affects product innovation and brand strategies. ESMEs should build efficient capabilities to cater to international customers' requirements, develop cross-functional marketing processes specific to foreign markets, innovate routines and appeal to local clients while maintaining global brand identity. This approach enables them to effectively translate market-specific insights gained through digital platforms into actionable global product and brand strategies. For example, this can be achieved through platform data assetization, intelligent marketing systems and modular product platforms. Consequently, ESMEs should optimize their IMCs to develop a set of dynamic marketing skills and competencies through

responsive and efficient cross-functional market processes to adapt to the changing global market environments, competing products and distribution channels and customer needs.

Third, managers should carefully consider product attributes and brand positioning needs when selecting cross-border platforms that align with their target foreign markets. In particular, striking a balance between platform embedding breadth and depth is essential. Platforms are called “two-sided markets” or “multi-sided markets”, equipped with mature marketing tools and global resource networks to expand ESMEs’ international marketing channels and product international visibility through search and promotion functions. However, excessive diversification across multiple platforms complicates market information management, while over-reliance on a single platform constrains strategic autonomy. To optimize international competitiveness, our results indicate that when embedding breadth at a medium level, coupling AI with strategic agility significantly benefits brand upgrading. This finding suggests that ESMEs aiming to enhance their brand advantages should appropriately maintain cooperative relationships with multiple cross-border platforms. If they prefer to pursue product innovation advantages, ESMEs should prioritize establishing close collaborative relationships over diversification, as economies of scope provide limited applicability in platform-based new product development context.

6.4 Limitations and future research

While this study provides an initial step in understanding how AI coupling with strategic agility should be organized within platforms and how it drives international value creation, this research has certain limitations that offer avenues for future research. Our findings are based on data from SMEs in China. While this region provides a relevant and timely context due to its proactive stance on digital platform use in internationalization, the results may not be fully generalizable to other geographical settings or international models. Future studies could test the robustness of our framework and compare cross-national contexts to assess how regulatory, cultural and infrastructural variations influence the effects of AI-enabled strategic agility and SMEs’ international competitive advantage.

In addition, while we recognized platform embedding characteristics by studying the moderating role of embedding breadth and depth, other characteristics of digital platforms may also affect their outcomes in internationalization. This suggests fruitful opportunities to examine additional platform characteristics, particularly typological variations such as sales, interaction and innovation platforms operating in domestic versus foreign contexts. Specifically, future work should therefore investigate how SMEs strategically deploy distinct platform types to achieve specific objectives, particularly investigating how interaction platforms drive brand upgrading while sales platforms facilitate product innovation.

Finally, self-reported data from SME managers regarding platform AI adoption and agility perceptions risks introducing bias. Future studies should incorporate secondary data for objective measures to strengthen the robustness and reliability of the results. Addressing this limitation would advance understanding of how cross-border platforms enhance SMEs’ post-entry competitive advantage across diverse contexts and conditions. Furthermore, the evolving nature of AI technologies introduces temporal limitations, as current platform tools and implementation strategies may soon become obsolete. Future scholars might investigate longitudinal studies to track the evolution of AI infrastructures and their strategic integration over time, assessing whether their impacts on marketing and innovation remain stable or fluctuate in response to technological advancements and market developments.

7. Conclusion

To conclude, our research investigates how cross-border platforms serve as catalysts for ESMEs’ upgrading within GVCs. We address the underexplored coupling role of AI adoption and strategic agility, and its impact on product innovation and brand upgrading. These effects

vary across platform embedding characteristics, depending on the breadth and depth of ESMEs embedded in the platform. Additionally, drawing on the dynamic capability hierarchy as a lens for interaction between platforms and ESMEs, we establish IMCs' mediating function in how the coupling of AI adoption and strategic agility shapes product innovation and brand upgrading advantage. Our arguments and findings, therefore, suggest that the deployment of AI infrastructure and incorporation strategy agility can be valuable instruments for platform owners to strengthen complementary international competitiveness, including product innovation and brand upgrading. Overall, as digital platforms become increasingly critical channels for ESMEs participating in the GVC, we hope this study will inspire further research into the relationship between digital platform usage in international marketing and acquisition of international advantages.

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